Quantum Mechanics in Solids

Unit-5

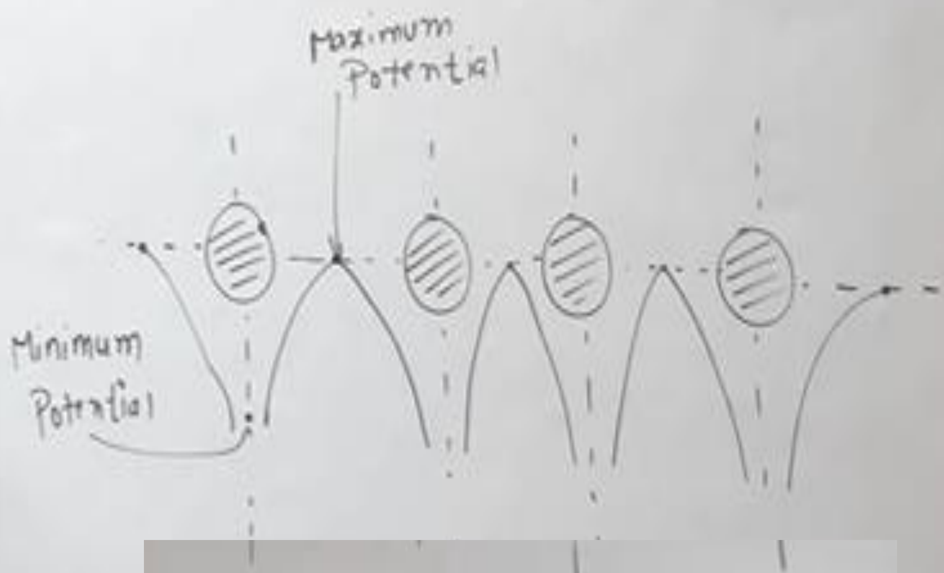
Lecture-5

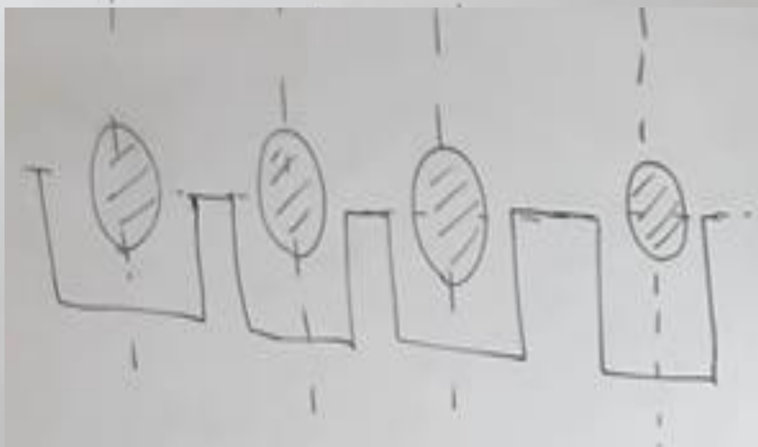
Bloch's theorem for particles in a periodic potential

Kronig-Penney Model

In Actual case constant Potential is replaced by periodic varying Potential

Consider I-D crystal





Kronig-Penney Model

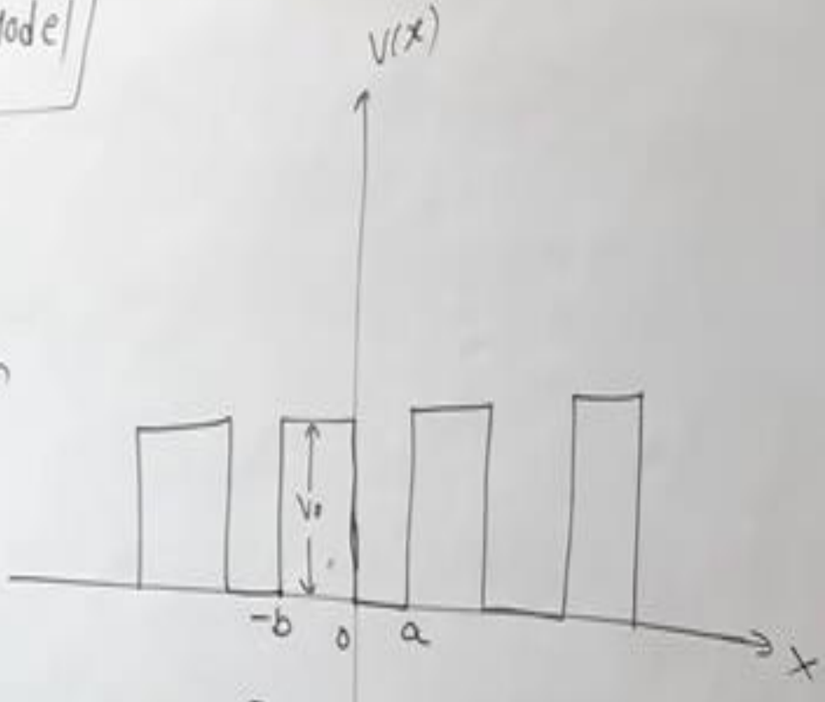
Basic Assumptions

$$V(x) = 0 \quad , \text{if} \quad 0 < x < a$$

Electron is inside the atom

$$V(x) = V_0 \quad , \text{if} \quad -b < x < 0$$

Electron is outside the atom



Periodic Potential

$$V(x+a+b) = V(x)$$

Kronig-Penney Model

Schrodinger Eq^n

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} E\psi = 0, \quad 0 < x < a \quad \text{— (1)}$$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_0)\psi = 0, \quad -b < x < 0 \quad \text{— (2)}$$

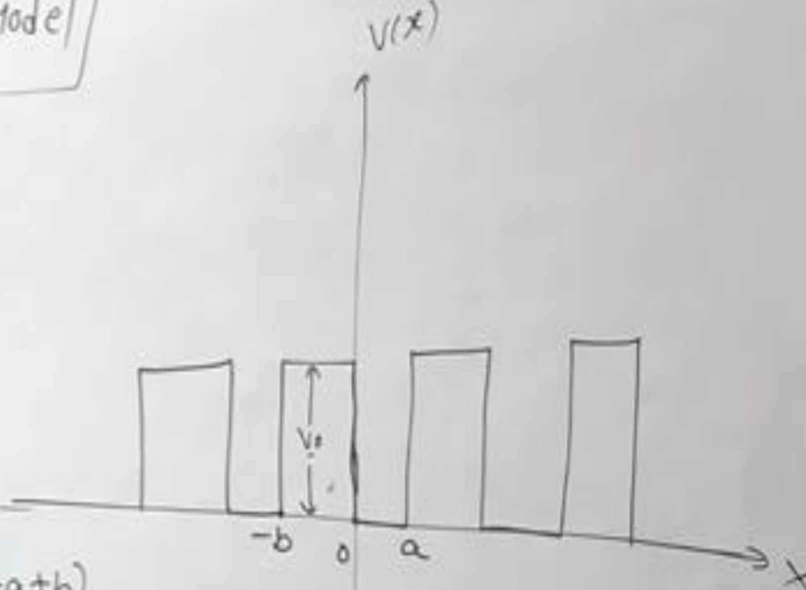
Let $\frac{2mE}{\hbar^2} = \alpha^2 \qquad \frac{2m}{\hbar^2}(V_0 - E) = \beta^2$

$$\frac{d^2\psi}{dx^2} + \alpha^2\psi = 0, \quad 0 < x < a \quad \text{— (3)}$$

$$\frac{d^2\psi}{dx^2} - \beta^2\psi = 0, \quad -b < x < 0 \quad \text{— (4)}$$

According to Bloch Theorem

$$\psi(x) = e^{ikx}\underbrace{u(x)}_{\text{Bloch Function}} \quad \text{— (5)}$$



Periodic Potential

$$u(x) = u(x + a + b)$$

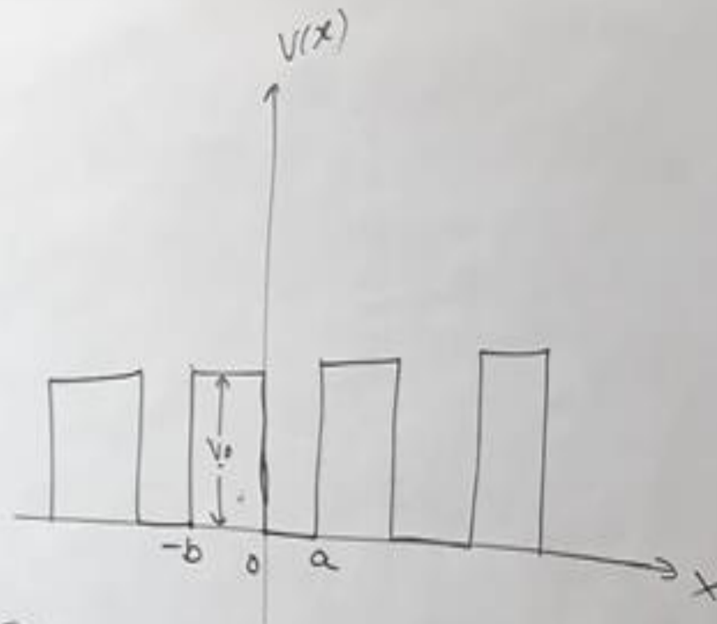
$$\frac{d\psi}{dx} = e^{ikx}\frac{du}{dx} + ike^{ikx}u$$

$$\frac{d^2\psi}{dx^2} = ike^{ikx}\frac{du}{dx} + e^{ikx}\frac{d^2u}{dx^2} + i^2k^2e^{ikx}u + ike^{ikx}\frac{du}{dx}$$

$$= e^{ikx}\frac{d^2u}{dx^2} + 2ike^{ikx}\frac{du}{dx} - k^2e^{ikx}u$$

$$\frac{d^2u}{dx^2} + 2ik\frac{du}{dx} + (\alpha^2 - k^2)u = 0 \ , \quad 0<x<a \quad — (6)$$

$$\frac{d^2u}{dx^2} + 2ik\frac{du}{dx} - (\beta^2 + k^2)u = 0 \ , \quad -b<x<0 \quad — (7)$$



General solutions

$$u_1 = Ae^{i(\alpha-k)x} + Be^{-i(\alpha+k)x} \ , \quad 0<x<a \quad — (8)$$

$$u_2 = Ce^{(\beta-ik)x} + De^{-(\beta+ik)x} \ , \quad -b<x<0 \quad — (9)$$

Boundary conditions

$$\left.\begin{aligned} (u_1)_{x=0} &= (u_2)_{x=0} \\ \left(\frac{du_1}{dx}\right)_{x=0} &= \left(\frac{du_2}{dx}\right)_{x=0} \end{aligned}\right] — (11)$$

$$\left.\begin{aligned} (u_1)_{x=a} &= (u_2)_{x=-b} \\ \left(\frac{du_1}{dx}\right)_{x=a} &= \left(\frac{du_2}{dx}\right)_{x=-b} \end{aligned}\right] — (12)$$

$$A + B = C + D \quad — (13)$$

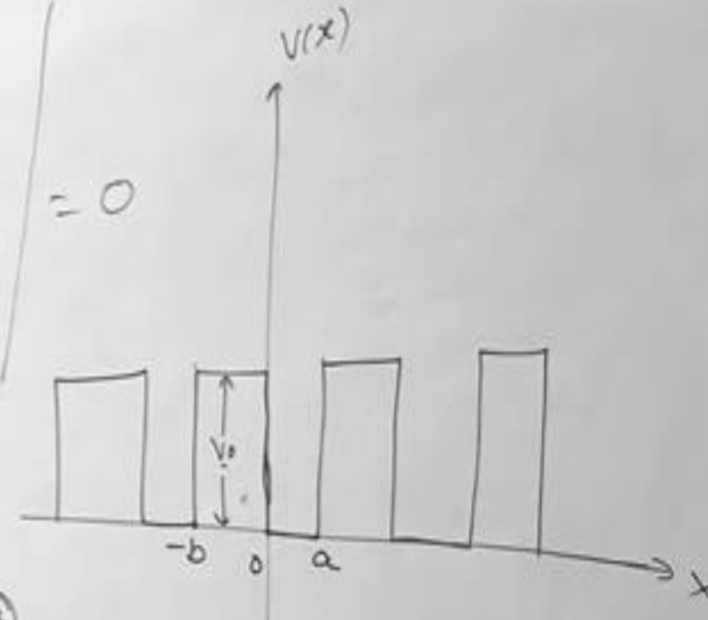
$$Ai(\alpha-k) - Bi(\alpha+k) = C(\beta-ik) - D(\beta+ik) \quad — (14)$$

$$Ae^{i(\alpha-k)a} + Be^{-i(\alpha+k)a} = Ce^{-(\beta-ik)b} + De^{(\beta+ik)b} \quad — (15)$$

$$Ai(\alpha-k)e^{i(\alpha-k)x} - Bi(\alpha+k)e^{i(\alpha+k)b} = C(\beta-ik)e^{-(\beta-ik)b} - D(\beta+ik)e^{(\beta+ik)b} \quad — (16)$$

Choose A, B, C & D

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ i(\alpha-k) & -i(\alpha+k) & (\beta-ik) & -(\beta-ik) \\ e^{i(\alpha-k)a} & e^{-i(\alpha+k)a} & e^{-(\beta-ik)b} & e^{(\beta+ik)b} \\ i(\alpha-k)\,e^{i(\alpha-k)x} & -i(\alpha+k)e^{i(\alpha+k)b} & (\beta-ik)e^{-(\beta-ik)b} & -(\beta+ik)e^{(\beta+ik)b} \end{vmatrix} = 0$$



$$\frac{\beta^2+\alpha^2}{2\alpha\beta}\sinh\beta b\,\sin\alpha a + \cosh\beta b\cos\alpha a = \cos k(a+b) \quad (17)$$

$V_0 \to \infty$, $b \to 0$

If $b \to 0$ $\quad \sinh\beta b \to \beta b$

$\cosh\beta b \to 1$

$$\frac{\beta^2+\alpha^2}{2\alpha\cancel{\beta}} \times \cancel{\beta} b\,\sin\alpha a + \cos\alpha a = \cos ka$$

$$\left(\frac{\beta^2+\alpha^2}{2\alpha}\right) b\sin\alpha a + \cos\alpha a = \cos ka \quad (18)$$

Now, $\alpha^2 = \frac{2m}{\hbar^2}E$

$\beta^2 = \frac{2m}{\hbar^2}(V_0 - E)$

$$\alpha^2+\beta^2 = \cancel{\frac{2mE}{\hbar^2}} + \frac{2m}{\hbar^2}V_0 - \cancel{\frac{2m}{\hbar^2}E}$$

$$\alpha^2+\beta^2=\frac{2mV_0}{\hbar^2}$$

$$\frac{\alpha^2+\beta^2}{2\alpha}=\frac{2mV_0}{2\alpha\hbar^2}=\frac{mV_0}{\alpha\hbar^2}$$

$$\frac{mV_0 b}{\alpha\hbar^2}\sin\alpha a+\cos\alpha a=\cos ka$$

$$\frac{mV_0 ab}{\alpha a\hbar^2}\sin\alpha a+\cos\alpha a=\cos ka$$

$$\underbrace{\left(\frac{mV_0 ba}{\hbar^2}\right)}\frac{\sin\alpha a}{\alpha a}+\cos\alpha a=\cos ka$$

barrier strength

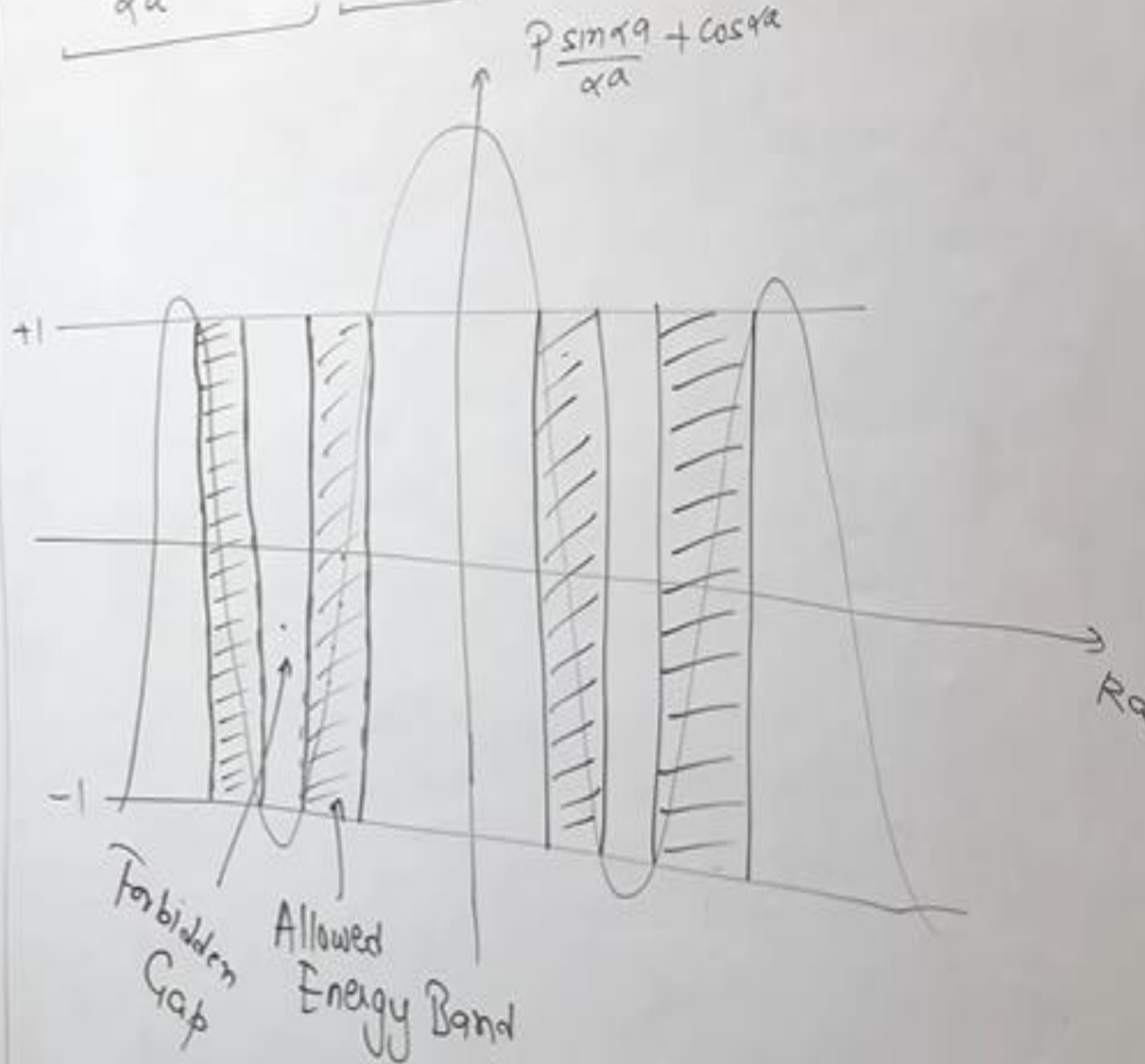
$$P=\frac{mV_0 ba}{\hbar^2}$$

$P\to\infty \Rightarrow$ Electron tightly bound

$P\to 0 \Rightarrow$ Electron is free

$$\underbrace{\frac{P\sin\alpha a}{\alpha a}+\cos\alpha a}=\underbrace{\cos ka} \quad \text{— (19)}$$

$\cos ka=\pm 1$



$$\alpha^2+\beta^2=\frac{2mV_0}{\hbar^2}$$

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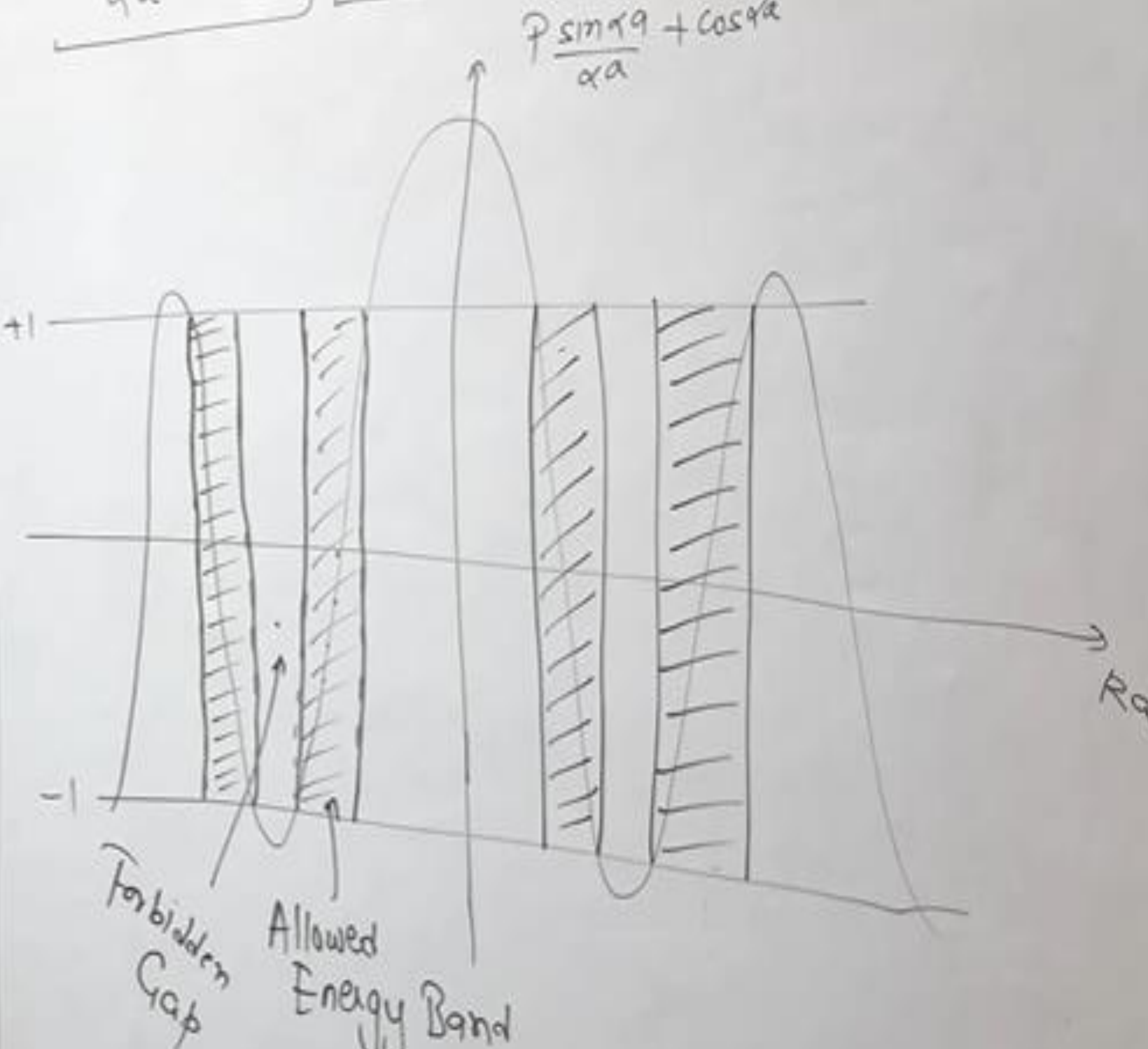


Conclusions:

① If $P \to \infty$, there is increase in barrier strength, width of allowed Energy bands become narrow

$$\sin \alpha a = 0$$

$$\sin \alpha a = \sin n\pi$$

$$\alpha a = n\pi$$

$$\alpha = \frac{n\pi}{a}$$

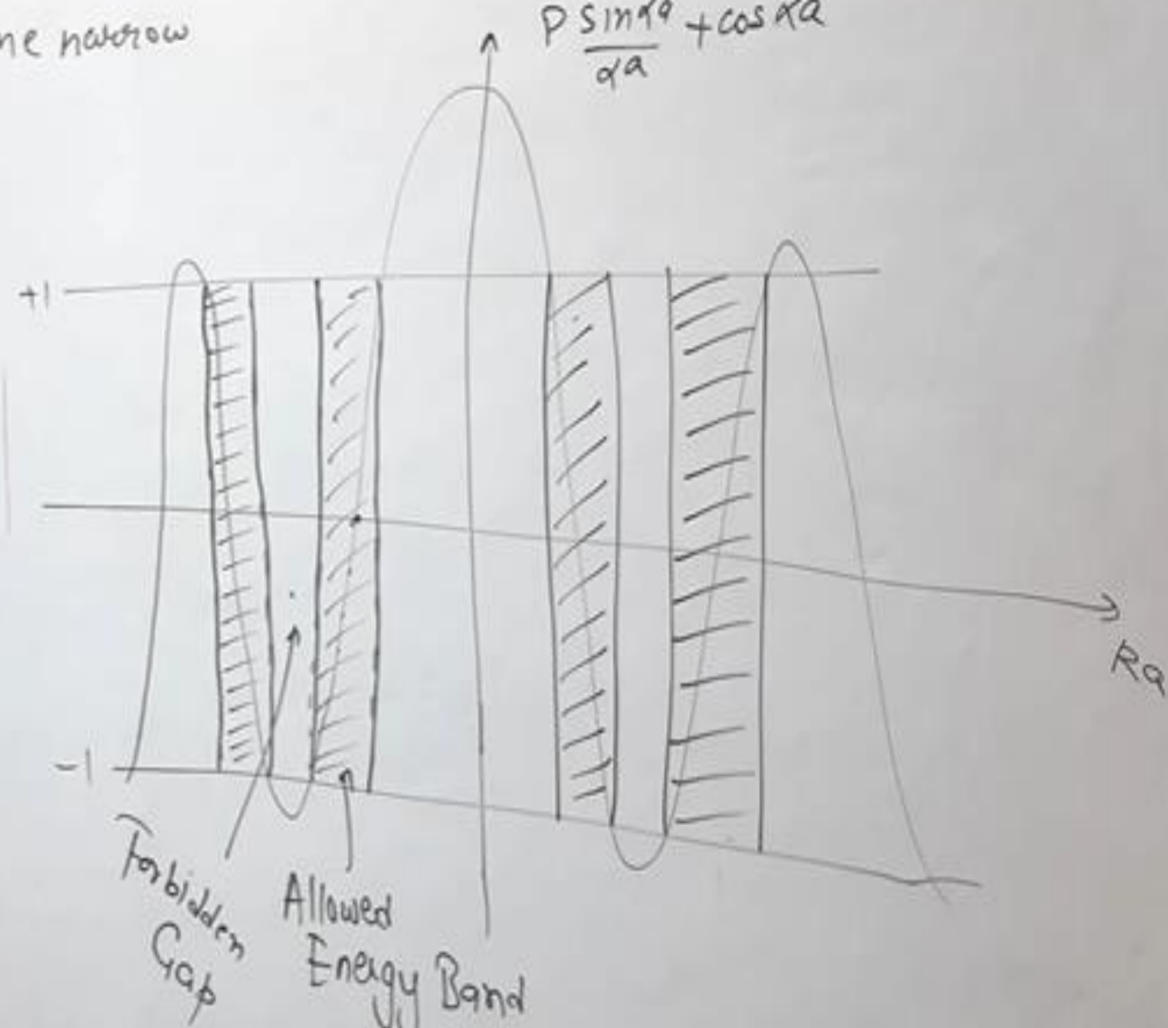
$$\alpha^2 = \frac{n^2\pi^2}{a^2}$$

$$\frac{2m}{\hbar^2}E = \frac{n^2\pi^2}{a^2}$$

$$\boxed{E = \frac{\hbar^2}{2m}\left(\frac{n\pi}{a}\right)^2}$$

Energy of bound Electron

lattice constant



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Energy of bound Electron

lattice constant

② If $P \to 0$

$$\cos \alpha a = \cos ka$$

$$\alpha a = ka$$

$$\alpha = k$$

$$\alpha^2 = k^2$$

$$\frac{2m}{\hbar^2} E = \left(\frac{2\pi}{\lambda}\right)^2 = \frac{4\pi^2}{\lambda^2}$$

$$E = \frac{\hbar^2}{2m} \times \frac{4\pi^2}{\lambda^2}$$

$$E = \frac{h^2}{\cancel{4\pi^2} \times 2m} \times \frac{\cancel{4\pi^2}}{\lambda^2}$$

$$E = \frac{1}{2m}\left(\frac{h}{\lambda}\right)^2$$

$$\boxed{E = \frac{P^2}{2m}}$$

Energy of free Electron

$$\left\{ \because P = \hbar k \quad P = \frac{h}{\lambda} \right\}$$

Crystal Momentum

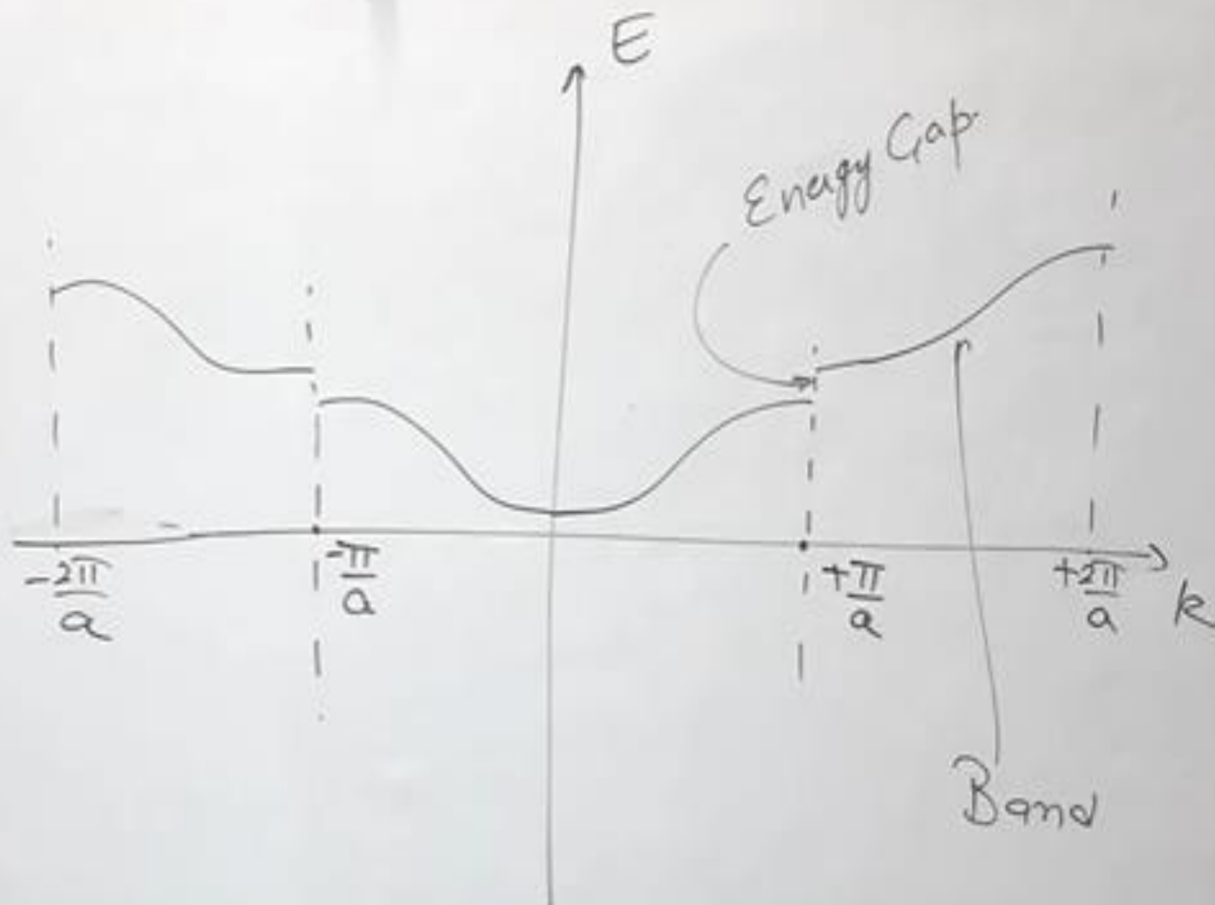
Conclusions:

③ Brillouin zones:

$\cos ka = \pm 1$

$ka = n\pi$

$$\boxed{k = \pm \frac{n\pi}{a}}$$



$k = \pm \frac{\pi}{a} \rightarrow$ First B.Z.

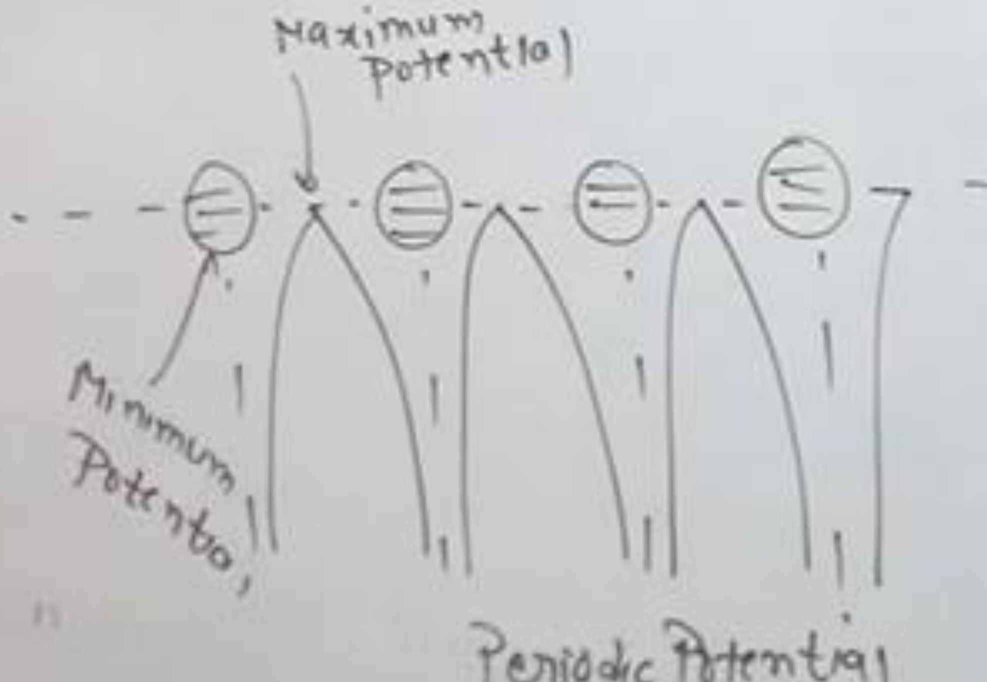
$k = \pm \frac{2\pi}{a} \rightarrow$ Second B.Z.

$$\boxed{E \propto k^2} \rightarrow \alpha a = ka$$

$$\alpha^2 = k^2$$

$$\frac{2m}{\hbar^2} E = k^2$$

Effective Mass:



$m_e = 9.1 \times 10^{-31}\,kg$

* Effective Mass:

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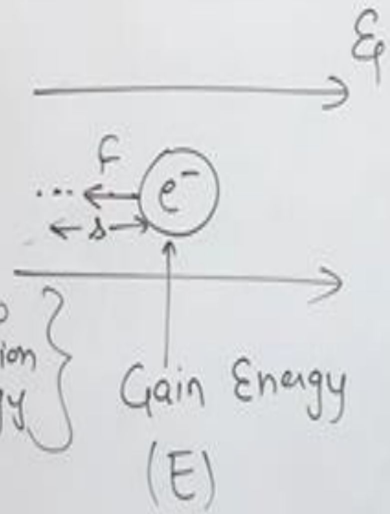

work done by Electron

$dw = F \cdot dS$

$dw = dE$ {∵ Acc^n to Conservation of Energy}

$dE = F\ dS.$

$$dE = F \cdot V dt$$

$$\cancel{dE} = F \cdot \frac{1}{\hbar} \frac{\cancel{dE}}{dk} dt$$

$$\frac{dK}{dt} = \frac{F}{\hbar} = \frac{e\mathcal{E}_g}{\hbar} \quad \text{— (1)}$$

$$a = \frac{dv}{dt}$$

$$a = \frac{d}{dt}\left(\frac{1}{\hbar}\frac{dE}{dk}\right)$$

$$a = \frac{1}{\hbar}\frac{d^2E}{dK^2} \times \frac{dK}{dt}$$

$$a = \frac{1}{\hbar}\frac{d^2E}{dK^2} \times \frac{e\mathcal{E}_g}{\hbar}$$

$$a = \frac{e\mathcal{E}_g}{\hbar^2}\frac{d^2E}{dK^2}$$

$$\frac{\cancel{e\mathcal{E}_g}}{m^*} = \frac{\cancel{e\mathcal{E}_g}}{\hbar^2}\frac{d^2E}{dK^2}$$

$$\boxed{m^* = \frac{\hbar^2}{d^2E/dk^2}}$$

Effective Mass